MIL-I-4715(MI)
7 June 1974
SUPERSEDING
MIS-15711
15 May 1966

MILITARY SPECIFICATION

INSULATING COMPOUND, ELECTRICAL EMBEDDING

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the properties of an aromatic amine cured epoxy resin which can be used for embedding or encapsulating electronic equipment of components. The cured compound is capable of operation continuously at 300 degrees Fahrenheit (F) with intermittant rises to 370 degrees Fahrenheit.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Federal

FED-STD-406	Plastics; Methods of Testing

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 5970

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:53Z
Check the source to verify that this is the current version before use.

Official Classification Committee

Uniform Freight Carrier Classification

(Application for copies should be addressed to the Official Classification Committee, One Park Avenue at 33rd Street, New York, New York, 10001).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. When specified by the procuring activity (See 6.2), a preproduction sample shall be prepared using the same methods proposed for subsequent production and shall be subjected to preproduction inspection (See 4.0).

3.2 Material. The material shall be a liquid epoxy resin to which is added an aromatic amine hardener.

3.3 Property requirements.

3.3.1 Specific gravity. The specific gravity of the cured compound shall be 1.20 plus or minus .05.

3.3.2 Heat distortion point. The heat distortion point of the cured compound shall be 275 degrees F minimum.

3.3.3 The weight change after submerging in boiling distilled water shall be minus 0.25 percent to plus 0.75 percent.

3.3.4 The weight change after submerging in boiling acetone shall be minus 0.25 percent to plus 0.50 percent.

3.3.5 Electrical insulation resistance. The electrical insulation resistance of the compound shall be 3×10^{12} ohms minimum at 95 degrees F and 95 percent relative humidity. After 100 hours under these conditions the electrical insulation resistance shall be 5×10^{9} ohms minimum.

3.3.6 Workmanship. The workmanship shall be such as to ensure a product which is uniform and in conformance with this specification. The resin and hardener shall be free of dirt, foreign material or other contaminents.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 202610-22T07:53Z
Check the source to verify that this is the currentversion before use.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herin, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification and tests. The testing of the material shall be classified as follows:

4.2.1 Preproduction inspection. (4.3).

4.2.2 Quality conformance inspection.(4.4).

4.3 Preproduction inspection. Preproduction inspection shall consist of all examinations and tests specified herein and shall be conducted only on preproduction samples. The preproduction samples shall be prepared using the same methods to be used for the manufacture of all subsequent production material.

4.4 Quality conformation inspection.

4.4.1 Quality conformance inspection tests shall be the examinations and tests required for each production lot. A production lot shall consist of all compounds of the same type, manufactured at one time.

4.4.2 Sampling. A random sample of sufficient quantity to complete the required testing shall be obtained from each production lot.

4.4.3 Quality conformance tests shall include the tests are requirements specified per 3.3.1, 3.3.2, and 3.3.6. Failure of any of the tests shall be cause for rejection of the production lot.

4.5 Test methods. All tests speciments shall be mixed and cured in accordance with manufacturer's instructions.

4.5.1 Specific gravity. To determine conformance to 3.3.1, the specific gravity shall be determined by testing in accordance with Method 5012 of FED-STD-406.

4.5.2 Heat distortion point. To determine conformance to 3.3.2, the heat distortion point shall be determined by testing in accordance with Method 2011 of FED-STD-406.

Source: https://assist.dla.mil -- Downloaded: 202610-22T07:53Z
Check the source to verify that this is the current version before use.

4.5.3 To determine conformance to 3.3.3, test in accordance with Method 7031 of FED-STD-406. Submerge specimens in boiling distilled water for 24 hours.

4.5.4 To determine conformance to 3.3.4, test in accordance with Method 7021 of FED-STD-406.

4.5.5 Electrical insulation resistance. To determine conformance with 3.3.5, the electrical insulation resistance shall be determined by testing in accordance with Method 4041 of FED-STD-406.

5. Preparation for delivery.

5.1 Packaging.

5.1.1 Level C. Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing.

5.2.1 Level C. The subject commodity packaged as specified herein shall be packed in accordance with the Uniform Freight Carrier Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 Marking. Marking shall be in accordance with MIL-STD-129 . and shall include but not necessarily be limited to the following:

a. This specification number.

b. Vendor's name.

c. Manufacturer's material designation.

d. Date of manufacture.

e. Quantity, size, etc.

f. Contract of procurement document number.

6. NOTES

6.1 Intended use. This compound will be used as a general purpose adhesive, encapsulant or casting resin.

Source: https://assist.dla.mil -- Downloaded: 202610-22T07:53Z
Check the source to verify that this is the currentversion before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether preproduction sample or test report is required.

c. Other special instructions.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-15711, dated 15 May 1966.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Project No. 5970-A447

☆U.S. GOVERNMENT PRINTING OFFICE: 1974—713-155/6372

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:53Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-22T07:53Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

DOD-314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Missile Command
ATTN: AMSMI-RCS
Redstone Arsenal, AL 35809

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:53Z
Check the source to verify that this is the current version before use.